

Perceval Beja-Battais

PhD Student in Applied Mathematics

Centre Borelli, ENS Paris-Saclay · in collaboration with Framatome

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Office 3S28, ENS Paris-Saclay, Gif-sur-Yvette 

Imitation Learning · Dynamical Systems · Machine Learning · Model Predictive Control

■ EDUCATION

PhD in Applied Mathematics

Apr 2024 – Mar 2027

Centre Borelli, ENS Paris-Saclay & Framatome — supervised by Nicolas Vayatis · Machine Learning for Optimal Control, Machine Learning for Simulation, Sensitivity Analysis, Physics-Informed ML, Safe ML — with application to the nuclear industry.

MVA Master (Mathematics, Vision, Learning) — Highest Honours

2022 – 2023

ENS Paris-Saclay · Statistical Learning, Deep Learning, Machine Learning, Computer Vision.

Mathematics of Modeling Master

2021 – 2022

Sorbonne Université · Optimization, Optimal Transport, Functional Analysis, Game Theory, Statistics, PDE.

Ingénieur Civil des Mines

2019 – 2022

Mines Nancy · Probabilities, Statistics, PDE, Optimization, Machine Learning and Deep Learning.

MPSI – MP

2017 – 2019

Lycée Lakanal · Two intensive years preparing the entrance exams of top French engineering schools.

■ PUBLICATIONS

P. Beja-Battais, A. Grossetête, N. Vayatis. **Structured Representation Learning for Behavior Cloning: How can we learn to safely control a nuclear power plant?** *NeurIPS 2026 Workshop AI4PowerGrids* [arXiv:2610.06211](https://arxiv.org/abs/2610.06211)

P. Beja-Battais, G. Dupré, A. Grossetête, N. Vayatis. **Accelerating Full-Scale Nonlinear Model Predictive Control via Surrogate Dynamics Optimization.** *Preprint, 2026* [arXiv:2604.05566](https://arxiv.org/abs/2604.05566)

P. Beja-Battais, A. Grossetête, N. Vayatis. **Enhancing Nuclear Reactor Core Simulation through Data-Based Surrogate Models.** *International Congress on Advances in Nuclear Power Plants (ICAPP), 2025* [hal-05360255](https://hal.archives-ouvertes.fr/hal-05360255)

G. Serré, P. Beja-Battais, S. Chirrane, A. Kalogeratos, N. Vayatis. **LIPO+: Frugal Global Optimization for Lipschitz Functions.** *SETN 2024* [arXiv:2406.19723](https://arxiv.org/abs/2406.19723)

P. Beja-Battais. **Overview of AdaBoost: Reconciling its views to better understand its dynamics.** *Preprint, 2023* [arXiv:2310.18323](https://arxiv.org/abs/2310.18323)

Patent

P. Beja-Battais, A. Grossetête, N. Vayatis. **A novel warm-start method for Optimal Control of Nuclear Power Plants.** *Patent, 2025*

■ RESEARCH & WORK EXPERIENCE

Pre-Doctoral Researcher

Apr 2023 – Mar 2024

Centre Borelli, ENS Paris-Saclay · Research on Machine Learning topics.

R&D Intern

Apr – Aug 2022

Framatome, Paris La Défense · Implementation from scratch of an optimization solver inside a dynamical system controlled by MPC algorithms.

Data Analyst Intern

Jun – Aug 2021

Editec UK Ltd., London (UK) · Modeling & forecasting of lotto sales; analysis of players' behaviour to design targeted new games.

■ TALKS, POSTERS & AWARDS

Poster — Machine Learning for Optimal Control of Nuclear Power Plants

Sep 2026

Machine Learning Summer School (MLSS), Tübingen

Talk — Enhancing Nuclear Reactor Core Simulation through Data-Based Surrogate Models

2025

International Congress on Advances in Nuclear Power Plants (ICAPP)

1st Prize — RL Challenge Accenta

2024

Collège de France

■ TEACHING & SUPERVISION

Teaching Assistant — Introduction to Statistical Learning

2026 – 2027

MVA Master, ENS Paris-Saclay

Mock oral examiner for engineering school entrance exams

2021 – 2025

IPESUP (MP, 2025) · Lycée Janson-de-Sailly (BCPST-2A, 2022–2023) · Lycée Condorcet (MPSI, 2021–2023)

Research internship supervision

2026

Efficient Sampling of Trajectories for Online Finetuning of Surrogate Simulation Schemes (May – Oct 2026)

Singular Perturbation of Nonlinear DAEs (Mar – Jun 2026)